

1. (a) $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$, $\mu(A_n) = \infty$, $\bigcap_{n \geq 1} A_n = \emptyset$. So $\mu(\bigcap_{n \geq 1} A_n) \neq \lim_{n \rightarrow \infty} \mu(A_n)$. Hence **m.c.f.a** doesn't hold.
- (b) Take $\mu'(A) = \sum_{i \in A} \frac{1}{2^i}$. Then $\mu(\bigcap_{n \geq 1} A_n) = 0 = \lim_{n \rightarrow \infty} \frac{1}{2^{n-1}} = \lim_{n \rightarrow \infty} \mu(A_n)$. So m.c.f.a holds.
- (c) If $\exists n_0$ s.t $\mu(A_{n_0}) < \infty$, then m.c.f.a holds.
- (d) Always true.

2. Taking $F = \Omega \in \mathcal{F}$, we get $\Omega_0 = \Omega \cap \Omega_0 \in \mathcal{F}_0$.

If $A = F \cap \Omega_0 \in \mathcal{F}_0$ for some $F \in \mathcal{F}$, then $\Omega_0 \setminus A = F^c \cap \Omega_0 \in \mathcal{F}_0$, since $F^c \in \mathcal{F}$ (note that $\Omega_0 \setminus A$ is the complement of A in Ω_0).

If $\{A_i\} \subset \mathcal{F}_0$, then each $A_i = F_i \cap \Omega_0$, for some $\{F_i\} \subset \mathcal{F}$. In this case, $\cup_{i \geq 1} F_i \in \mathcal{F}$ and hence $\cup_{i \geq 1} A_i = (\cup_{i \geq 1} F_i) \cap \Omega_0 \in \mathcal{F}_0$. This proves that $\mathcal{F}_0$ is a **σ -algebra** of subsets of Ω_0 .

3. (a) $f = 1_A$, where A is a **non-Borel measurable set**.

- (b) $g = 1_A - 1_{A^c} = \begin{cases} 1; & \text{on } A \\ -1; & \text{on } A^c \end{cases}$ is not measurable, but $|f| \equiv 1$ is **Borel measurable** (constant functions are always measurable).

5. (a) Note that for $B \in \mathcal{B}(\mathbb{R})$,

$$T^{-1}(B) = \begin{cases} \emptyset; & \text{if } 5 \notin B \text{ and } 0 \notin B \\ (-\infty, 0]; & \text{if } 5 \in B \text{ and } 0 \notin B \\ (0, \infty); & \text{if } 5 \notin B \text{ and } 0 \in B \\ \mathbb{R}; & \text{if } 5, 0 \in B \end{cases}$$

Since $(-\infty, 0], (0, \infty) \notin \mathcal{F}_1$, then T is not **$(\mathcal{F}_1, \mathcal{F}_2)$ measurable**.

- (b) Note that $\{\epsilon_i : i \geq 1\}$ are measurable, and for each $i \geq 1$, Y_i is created by adding a constant to ϵ_i and hence Y_i is measurable. For each $n \geq 1$, $\hat{\beta}_n$ is created by multiplication, component-wise addition and division(multiplication) by a constant of measurable functions $\{Y_i : i \geq 1\}$ and hence they are measurable too.